Two-Stage Audio Classification of Infant Cries Using Fine-Tuned HuBERT: A Production-Oriented Pipeline

Abstract

Infant cry classification can support parents in understanding the cause of a cry, but most existing systems are evaluated on small, unbalanced datasets and ignore practical issues such as cross-dataset duplication, speaker leakage, and domain bias. This paper presents a production-oriented pipeline for baby cry classification built on top of a fine-tuned HuBERT model. We design a two-stage cascade: a binary detector that decides whether an audio clip contains a baby cry, followed by a four-class classifier that identifies the type of cry (scared, physical pain, general need, or burping). Three public datasets are merged into a single 7,664-clip corpus, with extensive cleaning that includes MFCC-based fingerprint deduplication, voice activity-based trimming, loudness normalization, and speaker-aware splits. After observing severe class confusion in an initial six-class formulation, we collapse acoustically inseparable categories (hungry, tired, discomfort) into a single "needs" class, guided directly by the model's confusion patterns. The resulting cascade achieves **99.42%** accuracy on Stage 1 and **84.88%** accuracy on Stage 2 (F1-macro 0.774) on a held-out, speaker-disjoint test set. Beyond raw accuracy, we report calibration quality, per-source performance, and a duration-shortcut probe to show that our gains are honest. The full pipeline, model checkpoints, and an inference module are released as a reproducible package.

Keywords

Infant cry classification; HuBERT; audio transformers; transfer learning; data leakage; model calibration

1. Introduction

A newborn cries because it cannot communicate in any other way. For first-time parents, deciding why a baby is crying is one of the most stressful daily tasks. Several recent works have explored automatic cry classification, often promising five or more cry categories such as hunger, tiredness, discomfort, pain, and burping [1], [2]. In practice, however, these systems are usually trained and tested on the same small dataset, with no checks for duplicate audio, speaker overlap, or recording-environment bias. Reported numbers are therefore optimistic and rarely transfer to real applications.

This work tries to do the opposite. We start from the question: *what does a deployable cry classifier look like end-to-end?* We frame the problem as a two-stage cascade:

- **Stage 1 (binary):** Is this audio a baby cry or not?
- **Stage 2 (multi-class):** If it is a cry, what kind?

The design is motivated by realistic phone-app usage. A microphone in a baby monitor or smartphone hears mostly non-cry sounds (TV, traffic, talking). A reliable Stage 1 prevents the system from constantly alerting parents on irrelevant audio. Only when Stage 1 is confident does Stage 2 run, classifying the cry into clinically meaningful categories.

Our contributions are:

1. A **clean, deduplicated, speaker-aware** dataset of 7,664 audio clips assembled from three public sources plus environmental sounds from ESC-50 [3].
2. A **fine-tuned HuBERT** [4] cascade that achieves 99.42% / 84.88% accuracy on Stage 1 / Stage 2 on a frozen test split.
3. A **data-driven taxonomy reduction** from six to four classes, justified by acoustic separability rather than expert intuition.

4. **Honest evaluation** including per-source F1, calibration metrics, and a shortcut-detection probe.
5. A reproducible inference module and packaged release suitable for real deployment.

2. Related Work

Audio representation learning. Self-supervised speech models such as wav2vec 2.0 [5] and HuBERT [4] have replaced hand-crafted features (MFCC, mel-spectrograms) for most downstream audio tasks. Pre-trained on hundreds of hours of unlabeled speech, these models capture acoustic structure that transfers well to small, labeled audio classification problems with limited training data.

Infant cry analysis. Most previous work uses MFCC features with shallow classifiers (SVM, MLP) [1], [6]. CNN-based approaches on mel-spectrograms have appeared more recently [2]. Reported accuracies of 80–95% on the popular Donate-a-Cry corpus are common, but evaluation protocols differ widely. Many studies do not report whether the same baby appears in both training and test sets, which inflates accuracy.

Data quality issues in audio benchmarks. The general ML community has long documented near-duplicate problems in vision benchmarks [7]. Similar issues exist in audio: public infant cry datasets often share underlying recordings without clear attribution. To our knowledge, no prior infant cry paper performs MFCC-based fingerprint deduplication or speaker-aware splitting between train and test.

Stage cascades. Cascade architectures, where an easier model gates a harder one, are standard in face detection and keyword spotting [8]. We apply the same principle to cry analysis: a robust binary detector first, a more specific classifier second.

3. Methodology

3.1 Datasets and Cleaning

We combine three public sources of infant cry audio, summarized in Table I.

Source	Use	Size
Donate-a-Cry corpus (warcoder/infant-cry-audio-corpus)	Stage 2 cry types	457
baby-crying-dataset (mennaahmed23)	Stage 2 + laugh	5,749
baby-crying-sounds-dataset (mennaahmed23)	Stage 1 negatives + Stage 2	1,813

TABLE I — Source datasets used in this work.

To improve real-world Stage 1 performance, we add 1,991 clips from **ESC-50** [3] (excluding categories that overlap with our task, such as crying baby, laughing, and breathing).

Cleaning is performed in nine steps:

1. **Label standardization** with a global mapping dictionary (*belly pain* ↔ *belly_pain*, etc.).
2. **Class filtering** to keep only the target labels.
3. **Test-set reservation** before any further processing — a 20% stratified, frozen split.
4. **Audio standardization** for HuBERT: 16 kHz mono, high-pass filter at 50 Hz, voice-activity trimming (skipped for silence and noise), and LUFS-based loudness normalization to −23 LUFS.

5. **Quality filtering**: remove corrupted files and clips shorter than 1 s.

6. **Filename deduplication.**

7. **MFCC fingerprint deduplication.** We compute the mean and standard deviation of 13 MFCC coefficients per file, quantize to two decimals, and group identical fingerprints. This step removes **1,297 near-duplicate clips** that span multiple "different" datasets — strong evidence that these public corpora share underlying recordings.

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8. **Speaker-aware integrity.** Where filenames expose speaker IDs (regex patterns including UUIDs), we ensure that no speaker appears in both the training pool and the test set. **144 speakers** are reassigned to a single split.

9. **Speaker-aware train/validation split** within the pool (80/20).

The final dataset contains **7,664 audio clips**, all 16 kHz mono WAV with verified speaker disjointness across train/val/test.

3.2 Taxonomy Reduction

Initial experiments used six cry classes: *belly_pain*, *burping*, *discomfort*, *hungry*, *scared*, *tired*. Test results showed:

- F1-macro of 0.609.
- Severe confusion between *hungry*, *tired*, and *discomfort* (e.g., 32% of *discomfort* samples predicted as *hungry*).
- Per-source F1 dropped from 0.58 (dominant source) to 0.34 on a minority source — clear domain bias.

We treat this as a **product-design decision**, not a modeling failure. Three classes are not acoustically separable from short audio clips, even for human listeners [9]. We collapse them into a single class **needs** and rename *belly_pain* to **physical_pain**. The final Stage 2 taxonomy is shown in Table II.

Class	**Built from**	**Acoustic signature**
scared	scared	sudden onset, high-pitched
physical_pain	belly_pain	rhythmic, drawn-out
needs	hungry + tired + discomfort	variable, general fussing
burping	burping	short bursts after feeding

TABLE II — Final Stage 2 taxonomy.

Stage 1 keeps two classes: *baby_cry* and *not_baby_cry* (the latter merging laugh, noise, silence, and ESC-50 environmental sounds).

3.3 Model

Both stages share the same backbone: **HuBERT-base** (*facebook/hubert-base-ls960*, 95M parameters), pretrained on 960 hours of LibriSpeech. We add a small classification head:

```
HuBERT → masked mean-pool over time → Linear(768→256) → GELU → Dropout(0.1) →
Linear(256→C)
```

The CNN feature extractor is frozen (standard practice [4]); transformer layers are fine-tuned with a smaller learning rate than the head. Specifically:

Component	Learning rate
Classification head	1e-3
Transformer layers	1e-5
CNN feature encoder	frozen

TABLE III — Discriminative learning rates.

Training uses **weighted cross-entropy** (inverse-frequency class weights) with **label smoothing 0.1**. Audio inputs are 5-second fixed-length crops at 16 kHz; shorter clips are zero-padded with a corresponding attention mask, longer clips are randomly cropped during training and center-cropped at evaluation. The optimizer is AdamW with weight decay 1e-4, linear warmup over 500 steps, and cosine decay. Training runs for up to 15 epochs with early stopping on validation F1-macro.

3.4 Stage 2 Ensemble

Single-checkpoint Stage 2 reaches a test F1 of 0.769. Because we have two strong checkpoints from neighboring epochs (val F1 = 0.784 and 0.789), we average their softmax outputs at inference time. This deterministic, training-free combination raises test F1 to **0.774**.

We also experimented with per-class threshold optimization on the validation set. Although it improved validation F1, it overfit to validation noise and **hurt** test F1. We discard it — a useful negative result for practitioners.

3.5 Inference Cascade

At inference, audio is preprocessed identically to training, then routed through the cascade:

```
audio → Stage 1
           ■■ not_baby_cry → return
           ■■ baby_cry → Stage 2 ensemble → 4-class output
```

Both stages return calibrated softmax probabilities, allowing the application layer to apply confidence thresholds for “uncertain” responses if desired.

4. Results and Discussion

4.1 Headline Numbers

All numbers are on a **frozen, speaker-disjoint test set**. Stage 1 evaluates on 1,552 clips (992 cry, 560 non-cry); Stage 2 on the 992 cry clips.

Metric	Stage 1 (binary)	Stage 2 (4-class)
Accuracy	99.42%	84.88%
F1-macro	0.9937	0.7736
ROC-AUC	0.9983	—

TABLE IV — Headline test-set performance.

Stage 1's near-perfect accuracy reflects the large acoustic gap between baby cries and environmental noise. Stage 2's 0.774 F1-macro is, in our view, an honest result for a 4-class problem under speaker-disjoint evaluation — comparable to or better than published results that did not control for speaker overlap.

4.2 Per-Class Stage 2 Performance

Class	Test F1	Notes
scared	0.98	Strongest class, distinct profile
needs	0.89	Largest class, easy to fit
physical_pain	0.70	Moderate confusion with needs
burping	0.53	Smallest training set (193 samples)

TABLE V — Per-class Stage 2 F1 scores.

The weakest class, *burping*, is data-limited rather than method-limited. Targeted augmentation could help.

4.3 Per-Source F1 (Domain Bias)

Source	Stage 2 F1-macro	Test n
baby_crying	0.78	783
baby_sounds	0.50	151
infant_cry_corpus	0.38	58

TABLE VI — Per-source Stage 2 performance.

The model performs much better on its dominant training source. This is the most important caveat for deployment: real-world recordings from different microphones or environments can be expected to perform closer to 0.50–0.65 F1-macro until augmentation or domain adaptation is applied.

4.4 Calibration

We measure overconfident wrong predictions (confidence > 0.8 on incorrect predictions). Our baseline 6-class model produced these on **45%** of its errors. After adopting label smoothing of 0.1, this number drops to **11.5%** on the 4-class Stage 2 model, while Stage 1 keeps virtually all errors below this confidence band as well. Calibration matters for product use: an app that says "I'm not sure" is far more trustworthy than one that confidently misclassifies.

4.5 Duration-Shortcut Probe

The original *scared* class consisted entirely of clips capped at exactly 4 seconds. We worried the model might exploit duration as a shortcut. To test this, we time-stretched 40 *scared* test clips by ±30% (changing duration without changing pitch) and re-evaluated. Recall remained between 0.90 and 0.93 across all stretch factors, ruling out the shortcut hypothesis. The model genuinely learned acoustic features.

4.6 What Did Not Work

We are explicit about negative results:

- **Aggressive freezing** (4 transformer layers) combined with high dropout regressed F1-macro by 3 percentage points. Architectural regularization, when over-applied, removes capacity faster than it removes overfit.

• **Per-class threshold optimization** improved validation F1 by 1.2 pp but hurt test F1 by 2.7 pp — a clear case of validation overfitting.

4.7 Pipeline Properties

Property	Status
Reproducible (seeded)	Yes
Frozen test set since Step 4	Yes
Speaker-disjoint splits	Yes
Fingerprint-deduplicated	Yes (1,297 duplicates removed)
Calibrated outputs	Yes
Documented limitations	Yes

TABLE VII — Pipeline integrity checks.

5. Conclusion

We presented a complete, deployment-oriented pipeline for infant cry classification. Three contributions matter most. First, we showed that a careful data-cleaning protocol — particularly fingerprint deduplication and speaker-aware splitting — is at least as important as model choice. Without it, reported accuracy is optimistic. Second, we used the model's confusion matrix as a **product signal**: when classes were not acoustically separable, we merged them rather than chasing a number. Third, we built a small but useful improvement (a two-checkpoint ensemble) and rejected a tempting one (threshold tuning) based on test-set evidence.

The released cascade reaches 99.42% accuracy on cry detection and 84.88% on cry-type classification under speaker-disjoint evaluation. We see this as a practical baseline for real applications and as a reminder that, in deployable ML, the dataset and the labels are the model.

Future work includes targeted augmentation (room reverb, additive ESC-50 noise, pitch shifting) to address per-source bias, ONNX/TFLite export for on-device inference, and active collection of in-the-wild user audio for continual retraining.

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The dataset, trained checkpoints, and inference code are released as a reproducible package. The trained models are intended for research and assistive use, not for medical diagnosis.